ENGINEERING REVIEW · 61 TICKETS · 5 REVIEWERS · 11 NOV 2026

Holdfast — what the review found

Five AI reviewers (Agent Orchestrator) checked every ticket for money safety, race conditions, PayPal limits, dependencies and timing. This is the plain-language version of `docs/review/SUMMARY.md`.

TICKETS REVIEWED	BLOCKERS	MAJOR ISSUES	DESIGN RULES TO AGREE
61	14	~40	19
HF-001 to HF-061	after removing duplicates	grouped below	fix most blockers at once

> **Your decisions (11 Nov):** keep **every feature** — nothing is dropped. Build with **Agent Orchestrator** workers instead of a human team, so many tickets run in parallel. Use the existing PayPal sandbox credentials. Spigot comes after Holdfast.

01 The bottom line

- **The design is sound, but the money code has race conditions on paper.** Most of them are cheap to fix now (a few rules in the spec) and expensive once code exists.
- **There is no app code yet.** The very first tickets — project setup (HF-001) and the database schema (HF-002) — block everything else.
- **Freeze the database schema early.** Every money ticket depends on it; changing it mid-build costs hours across all workers.
- **Some PayPal paths are still unproven** (the in-page card form, PayPal-wallet holds, webhooks, a large payout). A 1-hour test pack (new ticket HF-063) should run first.
- **Timing:** with a human team the reviewers said P0 didn't fit (133 h vs ~76 h). With Agent Orchestrator running many workers in parallel, the limit becomes the **dependency chain**, not hours: the longest chain is about 15–18 hours of sequential work.

02 The 19 design rules (agree these before coding)

Each rule answers several reviewer findings and spec questions.

#	RULE	WHAT IT MEANS IN PLAIN WORDS
D1	**One door for every hold**	Every new PayPal hold (card, saved card, wallet, raise, re-bid) goes through one function. It re-checks that the sale is still open and the shopper's limit is fine, then marks the bid live — or immediately releases the new hold if anything changed.
D2	**Clear, Cancel and Expire lock the same way**	They lock the listing first, then the bids, then act. If anything moved underneath, they stop and ask again — never half-finish.
D3	**One live bid per shopper per listing**	A second bid is refused ("you already have a bid — raise it"). Lowering = cancel and re-bid (you lose your place in line).

D4	**Every release has one fixed ID**	`void-<bidId>` everywhere, so a hold can never be released twice or in conflicting ways.
D5	**Every PayPal call has a fixed duplicate-safety ID**	Create, authorize, charge, release, payout and tracking each use a set ID, so retries never double-charge.
D6	**The clock is absolute**	Hold age counts from PayPal's own authorization time. Rounds end at most 6 h after bidding closes; Clear must happen within 60 h of opening.
D7	**Who can bid when**	New bids only while bidding is open. Starting live rounds closes new bidding; during rounds, only raises are allowed.
D8	**No hold, no win**	A bid without a real PayPal hold can never reach Clear. The demo seed stops if more than 10% of holds fail.
D9	**Backups are promoted immediately**	When a winner's card fails, the next bidder is promoted in the same step, so the brand isn't paid before the backup is charged.
D10	**Backup list rules**	Everyone above the floor stays in reserve until all charges finish; below-floor bids are released at once. Max 3 backups per unit. Skip a backup whose hold is too old, whose owner paused, or who'd now pay above the store price.
D11	**Retries**	Up to 4 tries (1 s, 4 s, 16 s). Errors are sorted into "try again", "final" and "check PayPal first". Stuck jobs are taken over after 60 s.
D12	**Shopper limit counts every live hold**	Including holds still being approved, minus the one being replaced by a raise.
D13	**Easy sign-in for the demo**	Shoppers get an anonymous account on first visit; a demo-mode button signs in judges as shopper or brand.
D14	**Wording says "you pay your bid"**	"You won at your bid of $229. Charged $229." Releases add "your bank may show it as pending for a few days."
D15	**Demo brand is Aurel Audio**	No real brands in the video; real Channel3 data still shown in search and price comparisons.
D16	**Designs come from Figma "Holdfast — v3"**	Build screens from v3 components without waiting for every frame.
D17	**Price-watch subscription per product**	Stored on the product (and variant), not the listing; brands list allowed store domains.
D18	**Notifications start in-app**	Bid timeline, error states, MCP and 5-second polling during rounds first; email and push follow.
D19	**Reconcile with PayPal**	A job asks PayPal about every uncharged hold and releases any that shouldn't exist. The demo reset uses it before wiping data.

03 The 14 blockers, in plain words

#	WHAT COULD GO WRONG	FIX
B1	`DOUBLE CHARGE` A raise racing Clear, or two tabs, could leave one shopper with two winning bids.	Rules D1–D3: atomic swap, one live bid per shopper, Clear keeps only each shopper's highest bid.
B2	`FORGOTTEN HOLD` A hold that lands just after Clear/Cancel is never charged or released.	D1: re-check under the listing lock and release the new hold if the sale moved on.
B3	`CLEAR READS STALE DATA` A cancel or price-drop release sneaks in between Clear reading and writing.	D2: lock first, and stop if anything changed.
B4	`RETRY LOOKS LIKE FAILURE` PayPal says "already charged" on a retry; we'd treat it as a decline and charge a backup too — two people pay for one item.	On any such error, ask PayPal for the hold's real state and act on that.
B5	`BRAND PAID TOO EARLY` The payout fires before a backup bidder has been charged.	D9: promote in the same step; "done" means every winner is actually charged.
B6	`STUCK SETTLEMENT` A crash leaves charges or a whole sale stuck forever.	60-second takeover lease plus a "to-do" row written with Clear, picked up every minute.
B7	`BIDS WITHOUT HOLDS WIN` They'd trigger a chain of failed charges and inflate the payout.	D8; payout counts only completed charges.
B8	`BROWSER CARD PATH UNPROVEN` Our tests used a card server-side; the real in-page card form works differently (may need 3-D Secure).	New two-step flow + test pack first. Fallback: a labelled "Use sandbox test card" button that is already proven.
B9	`LIMIT RACE` Two bids placed at once could both pass the shopper's spending limit.	D12, re-checked inside D1.
B10	`AGENT CAN CHANGE THE AMOUNT AFTER "YES"` The confirmation goes back through the AI, which could re-issue a different bid.	Store the exact pending action; "confirm" runs exactly what was stored.
B11	`HOLDS LEFT DANGLING` Declined winners, half-released bids, abandoned approvals — the nightly sweep missed them.	Sweep every 5 minutes covering all cases, plus D19 reconcile.
B12	`DEMO PRICE-DROP WOULD WIPE HALF THE BOOK` Most fake bids would be released, breaking the "Clear 240" beat.	Drop the price just under one real bid; fixed demo bid table with printed expected numbers.
B13	`HOSTED DEMO WOULD BE DEAD FOR JUDGES` The sale expires, holds age out, judges can't log in.	Daily automatic demo refresh until Dec 21, demo logins, all live before submission.
B14	`SCHEMA GAPS` Missing columns would force mid-build changes across every track.	Full column list added to HF-002; freeze it early.

04 Major issues, grouped

Money correctness

- Block charging more than the bid (PayPal allows up to 115%).
- Authorize calls need duplicate-safety IDs too.
- Duplicate-request keys must be per user and must not "remember" retryable failures.
- Payouts: retryable attempt IDs, read each item's status, pay only completed charges.
- Separate "retry" from "failed"; cancel during rounds allowed (runs in background).
- Mark a shopper's limit fulfilled after a successful charge.

Live rounds & concurrency

- Only one round ticker at a time; one nudge per shopper per round.
- Auto-raise is duplicate-safe and never above the store price when price-watch is on.
- Clear button handles "book changed" with a one-click re-confirm; "Stop rounds & clear" during rounds.
- Price watcher: loop until nothing left, compare with the price at bid time, reprocess missed events.

PayPal & platform reality

- New test pack HF-063 (1 h, first): card form, wallet, replays, double void/charge, ~$50k payout, saved-card holds, expired holds, 300 holds over 4–5 test cards.
- A fake-PayPal mode for race tests and dev data.
- Render Workflows unproven: in-process runner first, Workflows switched on once tested.
- PayPal webhooks, wallet holds and Log in with PayPal are unproven — each gets a fallback.

Integrations, demo, submission

- Channel3 offers have no barcode field — match on product, variant, condition, store and USD.
- Don't leak the brand's floor through "likely to win" hints.
- MCP fixes (cent amounts, "already bid" error, setup snippet); buyer-agent chats must not leak across shoppers.
- Sponsor claims must be true (AG Studio, APIMatic, Bryntum, Render Workflows shown).
- **Rotate the PayPal sandbox secret** before the repo goes public; scan history for secrets.
- Record safety takes as each demo beat works.

05 How the tickets change (nothing dropped)

The reviewers proposed cuts for a 4-person team. **Per your decision, every feature stays in scope.** Instead, tickets are split so Agent Orchestrator workers can run them in parallel, and ordered so the money path is proven first.

CHANGE	TICKETS
Split into smaller pieces	HF-003 (core helpers a/b), 012 (create vs cancel), 014 (bid core vs card/wallet flow), 017 (limit rules vs UI), 030 (in-process runner vs Render Workflows), 036 & 038 (rounds basic vs full auto-raise), 042 (replay vs live Channel3), 044 (4 MCP tools vs raise tool), 048 (deploy vs demo refresh), 050 (Devpost draft vs video), 051 (recommendation vs Ask panel), 059 (won page vs payouts/tracking)
Merged	HF-015 into 014b (wallet), 020 + 021 (confirmation + error screens), 039 + 040 (buyer agent)
New	HF-063 PayPal test pack (first)
Built later, not dropped	Log in with PayPal (demo login first), listing wizard, brand overview, PayPal webhooks, notifications (email/push), embed widget, admin screen, landing page, Bryntum timeline, mandate auto-roll — all after the money path works
Dissolved into others	HF-047 hardening tests become acceptance tests inside the tickets that create each race

06 The plan with Agent Orchestrator

Times in IST. Workers run in parallel; checkpoints are where you look before the next wave starts.

WHEN (IST)	WHAT
Now	Agree the 19 rules · apply ticket edits · start HF-001 setup, HF-002 schema and HF-063 PayPal test pack in parallel
Checkpoint 1	Schema frozen · "one door for holds" merged · PayPal test results in (pick card fallback if needed)
Wave 2	Clear, bid placing, offer page, console grid, PayPal client, Channel3 fixtures, first Render deploy — all in parallel
Checkpoint 2	A real sandbox hold from the deployed offer page · Clear works on test bids · Devpost draft
Wave 3	Settlement runner, payout, promotion, sweeper, live rounds, MCP, buyer agent, price-drop replay
Checkpoint 3	**Money path green in sandbox:** hold → rounds → Clear → charge/release → declined card promoted → payout
Wave 4	Everything else (the "later" list above), polish, demo refresh, Render Workflows
Nov 12, ~16:30	Feature freeze → record video → README → Devpost
Nov 12, 23:30	**Submit** (hard deadline Nov 13, 01:30 IST)

> **Circuit breakers:** card form fails → test-card button · money path not green at checkpoint 3 → all workers switch to fixing it · Render Workflows misbehaves → stay in-process and say so in the README.

07 Open calls — and what you decided

QUESTION	DECISION
In-app buyer chat vs MCP only	`KEEP BOTH` (no features dropped)
Card path	Build the real card form; keep the proven test-card button as fallback
Who owns live rounds	An orchestrator worker; full version, not lite
Log in with PayPal	Keep it; demo login ships first as a fallback
Team	Agent Orchestrator workers instead of 4 people
PayPal credentials	Same sandbox app (rotate the secret after the hackathon)
Spigot	After Holdfast, also with Agent Orchestrator